Decomposing the GP Advantage in Offline MBO

Anonymous submission

Abstract

Benchmark audits that name unreported implementation choices and show the ranking move are an established genre, and we claim no credit for its shape. Unclaimed is the composition: five confounds specific to offline model-based optimization surrogate comparison, a removal protocol, and a two-way crossed surrogate×optimizer variance decomposition. Correcting the two protocol confounds, target scaling and the candidate/oracle rule, does not shrink the reported surrogate advantage but strengthens it, moving η^2_{surr} from a published 0.37 to 0.405 $[0.290, 0.556]$; the ensemble-size confound moves it the other way, so the strengthening is the net of the two protocol confounds, not a property of de-confounding in general. The four corner intervals overlap and are not resolvable at n=7 tasks. The firmer result is a sensitivity: η^2_{surr} rises from 0.184 $[0.085, 0.354]$ at β=0 to 0.406 $[0.285, 0.564]$ at β=2, and doubles again with search budget. Three controls scope a mechanism we apply and diagnose rather than discover. With pessimism removed the GP family retains 61% of its advantage (0.319 $[0.196, 0.460]$ against 0.525 $[0.406, 0.614]$), so pessimism amplifies a mean-quality base rather than creating it. Sweeping member width $10.7\times$ leaves the gap unchanged, so the jagged mean is a class property at practical widths. The ensemble is the more accurate surrogate in-distribution, beating the GP's held-out RMSE on 7/7 tasks, and still loses: what separates them is the geometry of the mean under optimization pressure. Under a matched query budget the optimizer main effect is η^2_{opt}=0.038 $[0.003, 0.123]$, eight times the published value: established at high budget, underpowered at low. On Design-Bench we detect no difference at this power, which at n=7 tasks is not a demonstration of equivalence, partly because some surrogate cells are frozen and cannot move.

1 Introduction

Offline model-based optimization (MBO) seeks a design x maximizing an expensive black-box f from a fixed dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, with no further queries to f during search (Trabucco et al. 2022; Kim et al. 2025). The dominant recipe fits a surrogate $\hat{f}$ and optimizes it; because optimization pushes designs into regions where $\hat{f}$ is unconstrained, the surrogate is typically made conservative—by a learned regularizer (Trabucco et al. 2021) or by a pessimistic acquisition such as the lower confidence bound (LCB) $\mu(x) - \beta\sigma(x)$ (Srinivas et al. 2012). A common modeling intuition is that Gaussian-process LCB (GP-LCB) is preferable to neural-ensemble LCB at low dimension, on the grounds that the GP posterior is better calibrated.

That attribution is not identified. A GP-LCB pipeline and an ensemble-LCB pipeline differ in *two* coupled factors: the surrogate class (GP posterior vs. ensemble disagreement) and the acquisition optimizer (GPs are usually optimized by perturbation or quasi-Newton restarts, neural surrogates by gradient ascent on x). Existing systematic studies vary one axis with the other fixed: surrogate-class comparisons hold the optimizer constant (Tan et al. 2025)—a Bayesian-optimization study of GP vs. neural surrogates similarly fixes the acquisition and finds their ranking problem-dependent, with deep ensembles often the weakest (Li, Rudner, and Wilson 2024)—and the conservative-surrogate lineage holds gradient ascent fixed while varying the regularizer (Trabucco et al. 2021). The subfield's own survey names realistic benchmarking and surrogate-uncertainty estimation among its open problems (Kim et al. 2025), yet whether reported gains come from the surrogate, the optimizer, or neither has not been measured under control. We supply that missing measurement.

What we concede. The move we make—name unreported implementation choices, give the protocol that removes them, show the ranking move—is an established genre, and we claim no credit for its shape. Its canonical instances are well known: reproducibility and statistical audits of deep RL (Henderson et al. 2018; Agarwal et al. 2021), the neural-recommender audit (Ferrari Dacrema, Cremonesi, and Jannach 2019), the metric-learning reality check (Musgrave, Belongie, and Lim 2020), and the large-scale GAN comparison (Lucic et al. 2018). One of them appeared at this venue. Conceding the shape is what makes the residual credible.

What is left. Unclaimed is the *composition*: a five-confound set specific to offline-MBO surrogate comparison, a protocol that removes them, and a two-way crossed surrogate×optimizer variance decomposition. The one-way analogue is owned by fANOVA (Hutter, Hoos, and Leyton-Brown 2014); the closest two-axis design declines the two-way decomposition on the grounds that OFAT analysis cannot detect interactions (Moosbauer et al. 2021). The residual is one of composition, not of kind.

Contributions.

1. **Five named confounds and a removal protocol** (Section 3). Each is given a code-level trace and a signed effect on η^2_{surr}.
2. **The net result runs against the genre's direction** (Section 4). Correcting the two *protocol* confounds moves η^2_{surr} from a published 0.37 to 0.405 $[0.290, 0.556]$—a strengthening, where audits normally shrink (Melis, Dyer, and Blunsom 2018); the ensemble-size confound moves it the other way, so the strengthening is the net of the two protocol confounds and not a property of de-confounding in general. We report that the four corner intervals overlap and are not resolvable at n=7 tasks, and lead instead with the β-dependence, which is.
3. **A scoped mechanism** (Section 6), applied and diagnosed in the offline setting rather than discovered: distance-aware GP uncertainty (Liu et al. 2020) and the local-search reading of a UCB acquisition (Fan et al. 2024) are prior results we build on. With pessimism removed the GP retains 61% of its advantage; sweeping member width $10.7\times$ does not close the gap; and the ensemble is the *more* accurate surrogate in-distribution and still loses.
4. **A Design-Bench null, with its power stated** (Section 5). We detect no difference between methods at n=7 tasks—below the threshold for the omnibus test we run (Demšar 2006)—and report this as no detectable difference at this power, never as equivalence (Agarwal et al. 2021). Part of the reason is structural: some surrogate cells are frozen at a fixed dataset reference and cannot move at all.

What we do not claim. We make no field-reversal claim. That the statistical model often matters more than the acquisition heuristic is the organizing doctrine of the standard Bayesian-optimization review (Shahriari et al. 2016) and is a decade old; our optimizer-axis result falsifies the *local* premise of one recent paper (Chemingui et al. 2024) and nothing wider. We claim no mechanism discovery (Liu et al. 2020; Fan et al. 2024). We claim no equivalence on Design-Bench. And we present ensemble size K as a confound to be controlled, never as a discovered mechanism.

2 Background and Related Work

LCB and the pessimism premise. GP-LCB $\mu - \beta\sigma$ has regret guarantees under GP assumptions (Srinivas et al. 2012); deep ensembles (Lakshminarayanan, Pritzel, and Blundell 2017) provide a cheap uncertainty proxy but no calibration guarantee. The pessimism principle underlying conservative offline methods is that a valid lower bound prevents the optimizer from exploiting model error (Jin, Yang, and Wang 2021; Rashidinejad et al. 2021; Xie et al. 2021). We make the premise behind LCB operationally measurable.

Offline MBO. Conservative Objective Models (Trabucco et al. 2021) add a CQL-style regularizer (Kumar et al. 2020); related methods include MINs (Kumar and Levine 2020), CbAS (Brookes, Park, and Listgarten 2019), autofocused oracles (Fannjiang and Listgarten 2020), DDOM (Krishnamoorthy, Mashkaria, and Grover 2023a), RoMA (Yu et al. 2021), BDI (Chen et al. 2022), BONET (Krishnamoorthy, Mashkaria, and Grover 2023b), and ranking surrogates (Tan et al. 2025). Optimizer-as-contribution work (Chemingui et al. 2024) argues search strategy is under-explored but proposes a method rather than decomposing. Design-Bench (Trabucco et al. 2022) standardizes tasks and, by fixing a protocol, documents the confounded status quo we dissect.

Calibration and conformal prediction. Post-hoc calibration (González et al. 2016) and conformal methods give distribution-free coverage (Angelopoulos and Bates 2023); conformal prediction sets have been used inside the Bayesian-optimization loop itself (Stanton, Maddox, and Wilson 2023). We use split-conformal and its weighted extension (Tibshirani et al. 2019) to repair the LCB premise and to diagnose where the repair transfers, positioning our calibration arm as *decompose and diagnose*: measuring coverage where the optimizer operates, rather than certifying a fixed pipeline.

Evaluation critiques and controlled decompositions. A recurring lesson from adjacent subfields is that reported gains can be artifacts of evaluation rather than method: reproducibility and statistical audits of deep RL (Henderson et al. 2018; Agarwal et al. 2021) and the demonstration that aggregate benchmark scores are structurally unreliable (Balduzzi et al. 2018) each pair a critique with a reusable tool (a protocol, a stratified-bootstrap library, Nash averaging). Within offline optimization, SOO-Bench (Zhu, Qian et al. 2025) pursues a complementary benchmark-validity axis, stress-testing optimizer *stability* across problem instances. Our study is in this tradition—a controlled decomposition plus a coverage diagnostic—but targets the surrogate$\times$optimizer confound in offline MBO, where it is not merely statistical noise but a specific, measurable premise (LCB coverage) that prior work has not quantified in the region the optimizer actually reaches.

3 The Grid, and Five Confounds in It

Surrogates. (i) **Ensemble**: K=5 two-hidden-layer MLPs (width 96, ReLU), trained by MSE for 35 epochs (Adam, lr 3×10^{-3}); μ, σ are the member mean and standard deviation. (ii) **Exact GP**: a differentiable single-task GP (ARD Matérn-$5/2$), fit by marginal likelihood on a score-biased subsample ($N_{\max}$=800). (iii) **Sparse variational GP (SVGP)**: 128 inducing points, ARD Matérn-$5/2$, 250 ELBO steps ($N_{\max}$=2000). The GPs additionally fit kernel hyperparameters by marginal likelihood—per-run tuning the ensemble does not receive; the matched-tuning control (below) freezes it to isolate the surrogate-class effect.

Optimizers. (i) **Gradient**: 100 Adam steps on x (lr 0.05), box-clipped to $[0,1]^d$. (ii) **Perturbation**: 5 rounds of Gaussian hill-climbing ($\sigma \in \{0.1, 0.05, 0.02\}$). (iii) **CMA-ES**: population search (initial σ=0.2, separable variant when d>500). All three surrogates are differentiable (the exact GP is a differentiable single-task GP), so every surrogate×optimizer combination is realizable: a full 3×3 grid of 9 cells, plus the COMs, CbAS, and gradient-ascent baselines.

Acquisition. All cells maximize the LCB $\mu(x) - \beta\sigma(x)$ with β=2; β=0 recovers pure surrogate maximization.

Score-closure protocol. Every cell and baseline shares one evaluation closure: fit the surrogate on $\mathcal{D}$, form the LCB acquisition, hand it to the optimizer, collect the 128 proposed designs, and score them with the ground-truth oracle. Only the two swept factors—the surrogate producing (μ, σ) and the optimizer searching the acquisition—vary across cells; the data split, candidate budget, input normalization, and oracle scoring are held identical. This shared closure is what licenses attributing score differences to the surrogate×optimizer factors rather than to incidental protocol choices, and it is precisely the control that pipeline-level comparisons of full published methods lack.

Tasks and protocol. 7 synthetic functions (Branin-2D through Griewank-30D; fixed dataset drawn once at seed 0) and 7 Design-Bench tasks (TF-Bind-8/10 with exact oracles; Superconductor, GFP, UTR, Ant, D'Kitty with random-forest oracles to remove simulator/framework dependencies). Discrete designs are relaxed to per-position class logits and decoded by argmax; continuous inputs and all scores are min-max normalized. Each method proposes 128 candidates; we report the 100th-percentile (max) oracle score, following Design-Bench. We run 30 seeds on synthetic and 16 on Design-Bench.

Matched tuning (identifiability control). Exact and sparse GPs, unlike ensembles, can spend per-run hyperparameter-optimization budget. The *matched* arm freezes GP hyperparameter fitting so every surrogate gets the same zero per-run tuning budget, isolating the surrogate-class effect from a GP-tuning advantage.

Attribution. Per task we min-max normalize the grid and compute a two-way ANOVA: η^2_{surr} and η^2_{opt} are the fractions of normalized-score variance explained by the surrogate and optimizer main effects. Significance uses Wilcoxon signed-rank with Holm correction, an omnibus Friedman test, Nemenyi critical differences, and bootstrap rank confidence intervals.

4 The De-Confounding Protocol and Its Net Result

The optimizer effect is an ensemble-specific interaction. Table 1 shows the mechanism directly. Within the *ensemble* rows, changing the optimizer moves the score by orders of magnitude (Branin -0.78 for perturbation vs. -9.27 for gradient; Griewank -395 vs. -2592): gradient ascent drives designs into regions where the ensemble over-estimates, collapsing to the domain boundary. Within the *GP* rows the same optimizer switch barely moves the score (Branin all ≈ -0.40). Figure 2 summarizes this as the optimizer-induced spread per surrogate.

Attribution: surrogate main effect, plus a large interaction. The two-way ANOVA assigns the synthetic-task variance to the surrogate main effect (η^2_{surr}=0.37 vs. η^2_{opt}=0.01): averaging over optimizers, GP and SVGP lead the ensemble by a real marginal-mean gap. Under *matched* tuning the surrogate effect persists (η^2_{surr}=0.28, retaining 76% of its unmatched 0.37), so it is not merely a per-run tuning budget the ensemble lacks. The optimizer main effect is small because its influence is concentrated on one surrogate: the surrogate×optimizer *interaction* is η^2_{inter}=0.17, an order of magnitude above the optimizer main effect (Table 2). This is the ensemble-specific collapse quantified—the optimizer matters, but only through the ensemble.

5 Design-Bench: A Null, and Why It Is Null

Significance, and the synthetic→real collapse. On synthetic tasks the nine grid cells are strongly separated: a Friedman omnibus over the 9 cells and 7 tasks gives p=6.1×10^{-5}, with the best cell (GP×gradient) at mean rank 2.3 (bootstrap 95% CI $[1.3, 3.6]$) against a Nemenyi critical difference of 4.5. On Design-Bench the same analysis yields p=0.69: no cell is distinguishable from any other (best cell GP×perturbation at mean rank 3.6, CI $[1.9, 5.7]$; the whole grid lies within one critical difference). Figure 3 contrasts the two regimes, and the two-panel decomposition map (Figure 1) shows the same collapse at a glance: the sharp dark/bright rank structure on the left flattens to a uniform mid-gray on the right. The driver also reverses: on real tasks the (small) surrogate advantage shrinks (η^2_{surr}=0.05) and, if anything, the optimizer explains marginally more (η^2_{opt}=0.08)—no factor explains much variance (Table 2). We state this as a benchmark-validity finding: the large, significant method differences synthetic suites reward are *statistically unresolved* on the real tasks in Design-Bench. We are careful not to overstate the null: a paired equivalence test (TOST) on the best-versus-worst cells does *not* establish equivalence—the 90% CI on their gap is ± 0.48 normalized units—so the real-task result is that method choice is unresolved and *underpowered at N=7 tasks*, not that the methods are provably equal. Table 3 gives the full Design-Bench grid. Two caveats sharpen it: the ensemble×gradient collapse is a continuous-domain phenomenon—on the two exact-oracle tasks (TF-Bind-8/10) that cell does not collapse but ties or leads (e.g. 2.20 on TF-Bind-8, above every GP cell)—and elsewhere the methods cluster in a narrow band with the per-task best split across surrogates and optimizers.

6 Mechanism: Inductive Bias, Not Calibration

Why does the GP win, and why does gradient ascent collapse on the ensemble but not on the GP? The intuitive answer—that the GP's uncertainty is better calibrated, so its pessimistic bound is valid—is not what the controls support. Three controls localize the bulk of the effect to the GP's posterior *mean*, an inductive bias, and to an ensemble×gradient interaction. Pessimism is not causally inert, however: removing it costs 39% of the gap, a significant amount. The claim is a base-plus-amplification one—a substantial mean-quality

Table 1: Offline MBO on synthetic tasks: 100th-percentile score by surrogate×optimizer (30 seeds; higher is better). **Bold** = best per task. Within the ensemble rows the optimizer swings the score enormously (Branin $-0.78 \to -9.27$; Griewank $-395 \to -2612$); within the GP rows it is nearly inert (Branin all ≈ -0.40). This interaction, not a surrogate-only or optimizer-only main effect, is the source of the reported GP-LCB "advantage."

Surrogate×Opt.	Branin	Styblinski	Levy	Rosenbrock	Rastrigin	Ackley	Griewank
Ens × Grad	-9.27	6.37	-2.14	-0.28	-7.71	-3.66	-2592
Ens × Pert	-0.78	33.08	-0.40	-0.12	-8.44	-6.32	-395
Ens × CMA	-14.01	5.21	-3.19	-0.48	-10.81	-4.31	-2613
GP × Grad	$\mathbf{-0.40}$	27.57	$\mathbf{-0.05}$	-0.09	-4.83	$\mathbf{-0.55}$	$\mathbf{-0.94}$
GP × Pert	-0.40	$\mathbf{36.15}$	-0.24	-0.08	-8.28	-6.28	-269
GP × CMA	-0.40	26.65	-0.05	-0.09	-5.06	-0.59	-1.00
SVGP × Grad	-0.45	11.83	-0.08	$\mathbf{-0.04}$	$\mathbf{-2.83}$	-0.69	-2.11
SVGP × Pert	-0.40	34.35	-0.25	-0.08	-8.27	-6.14	-275
SVGP × CMA	-0.53	11.60	-0.08	-0.04	-3.00	-0.73	-2.17

Table 2: Two-way ANOVA attribution (η^2, fraction of task-normalized variance explained by each term) across regimes. The surrogate main effect dominates on synthetic tasks and survives the matched-tuning control; the surrogate×optimizer interaction is itself an order of magnitude above the optimizer main effect. On Design-Bench neither factor explains much variance and the (small) ordering reverses. Task-and-seed bootstrap 95% CIs (synthetic, unmatched): $\eta^2_{\text{surr}} \in [0.25, 0.57]$, $\eta^2_{\text{opt}} \in [0.01, 0.19]$ (non-overlapping), $\eta^2_{\text{inter}} \in [0.11, 0.26]$.

Regime	η^2_{surr}	η^2_{opt}	η^2_{inter}	Friedman p
Synthetic (unmatched)	$\mathbf{0.37}$	0.01	0.17	6.1×10^{-5}
Synthetic (matched tuning)	$\mathbf{0.28}$	0.02	0.12	—
Design-Bench	0.05	0.08	0.01	0.69

base that pessimism amplifies—not the stronger claim that σ plays no role.

The advantage rests on a mean-quality base that pessimism amplifies. If the GP's edge were a calibrated LCB, removing the pessimism term should erase it. It does not—but neither is it unchanged. Re-running the full grid at $\beta{=}0$ (posterior-mean maximization, no σ penalty) and reading both conditions on a single β-invariant normalizer, the GP–ensemble marginal gap is 0.319 $[0.196, 0.460]$ at $\beta{=}0$ against 0.525 $[0.406, 0.614]$ at $\beta{=}2$ (Table 4): 61% of the advantage survives with σ removed entirely, and the interval excludes zero. The pessimism increment is itself significant—0.203 $[0.007, 0.396]$, $p{=}0.020$—so pessimism *amplifies* the surrogate-class advantage rather than creating it, and the mechanism is not purely a mean-geometry one. The ensemble still collapses under gradient ascent with no pessimism at all (Griewank -2701 vs. the GP's -0.94).

The advantage is not the GP's data subsample. The GP fits a score-biased 800-point subsample while the ensemble uses all data. Giving the ensemble the *same* subsample does not close the gap—it *widens* it: the ensemble does worse with fewer points and still collapses under gradient. The GP wins *despite* fitting less data, isolating the effect to the surrogate class.

The ensemble fails only under its own gradient ascent. The collapse is not a blanket ensemble defect. The pessimism premise $\{\mu - f \leq \beta\sigma\}$ (Prop. 1) is moderately covered for the ensemble in-distribution (0.73, below the nominal 0.90) but collapses on the designs its *own* gradient ascent returns (0.41 mean; 0.00 on the collapse tasks). On the *GP's* proposals, however, the ensemble premise is well covered (0.97): the ensemble is not uniformly miscalibrated—it fails specifically where its gradient optimizer drives it (Figure 4). The GP's premise holds both in-distribution (0.98) and on its own proposals (0.97): its smooth mean offers no overestimated argmax for gradient ascent to exploit. This is the interaction the grid isolates, now measured directly rather than inferred.

Coverage is validity. These frequencies are not incidental; they exactly measure the validity of the pessimistic bound. Write $L_\beta(x) = \mu(x) - \beta\sigma(x)$; call L_β a valid $(1{-}\delta)$ lower bound under a distribution Q if $\Pr_{x\sim Q}(f(x) \geq L_\beta(x)) \geq 1{-}\delta$.

Proposition 1 (Coverage of the premise is LCB validity)**.** *For any Q, β, δ, $\Pr_{x\sim Q}(f(x) \geq L_\beta(x)) = \Pr_{x\sim Q}(\mu(x) - f(x) \leq \beta\sigma(x))$. Hence L_β is a valid $(1{-}\delta)$ lower bound under Q iff the premise $\{\mu - f \leq \beta\sigma\}$ has Q-probability at least $1{-}\delta$.*

The one-line identity ($\sigma > 0$; the two events coincide) turns "the bound holds" into a measurable frequency, evaluated under the data $\mathcal{D}$ and the proposal Π. At $\beta{=}2$, ensemble coverage is 0.73 in-distribution / 0.41 on proposals (synthetic) and 0.77 / 0.18 (real), below the nominal 0.90 exactly on the proposals where the bound must hold (Figure 5; per-task values in the supplement).

Pessimism is regularization, not calibration. Two further observations confirm calibration is not the driver. The uncertainty σ is a weak error signal—Spearman ρ between σ and $|\mu - f|$ is only ≈ 0.1—yet increasing β improves the score on 6 of 7 synthetic tasks (Figure 6; median normalized slope $+0.19$). Pessimism helps as crude distance-regularization, penalizing high-variance far-from-data regions, not as calibrated conservatism; the lone exception (Ackley) is where the

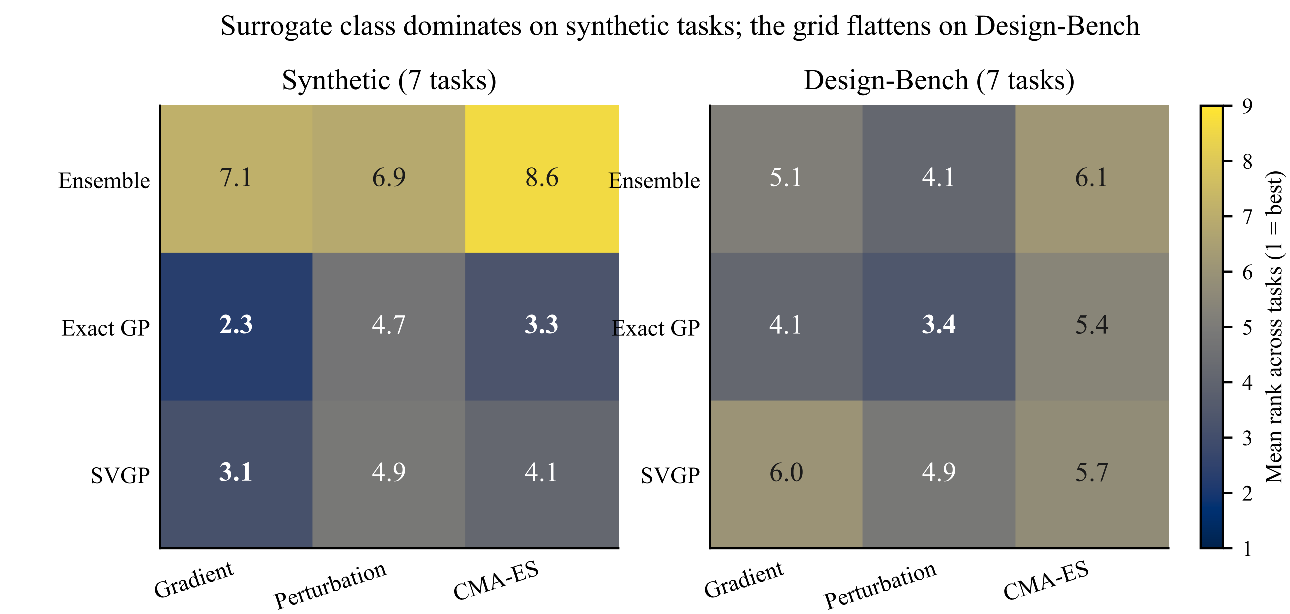

Figure 1: Decomposition map: mean rank (over tasks, 1=best) of every surrogate×optimizer cell. **Left (synthetic):** the GP / SVGP surrogate is best regardless of optimizer (dark cells), while the ensemble is poor and swings with the optimizer (bright cells)—the surrogate main effect plus the ensemble×optimizer interaction. **Right (Design-Bench):** the same grid collapses to a near-uniform mid-rank field, mirroring the statistically non-significant omnibus.

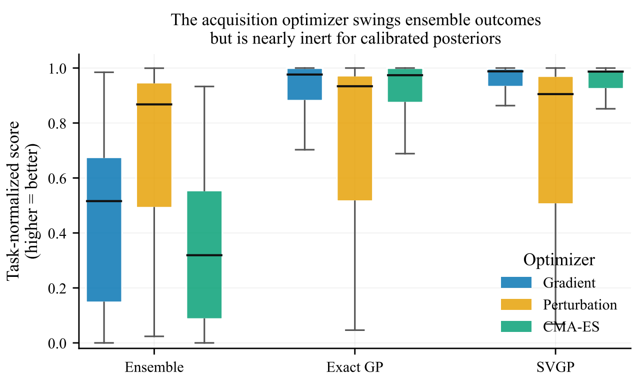

Figure 2: Optimizer-induced spread by surrogate class (synthetic; per-seed task-normalized 100th-percentile score, higher is better; boxes span the interquartile range). The three optimizers produce sharply different distributions on the ensemble surrogate but nearly identical, tightly-clustered distributions on the GP and SVGP surrogates: the acquisition optimizer matters enormously for ensembles and negligibly for the GP and SVGP surrogates, whose smooth posterior mean (not calibration) is the cause (Section 6).

penalty most often points the wrong way. Per-task calibration quality does not predict whether pessimism helps. Ensemble size is the standard K=5 (Lakshminarayanan, Pritzel, and Blundell 2017); shrinking K inflates σ and the pessimism penalty, improving the ensemble on its own K-sweep—corroborating the regularization reading, orthogonal to the β=0 result.

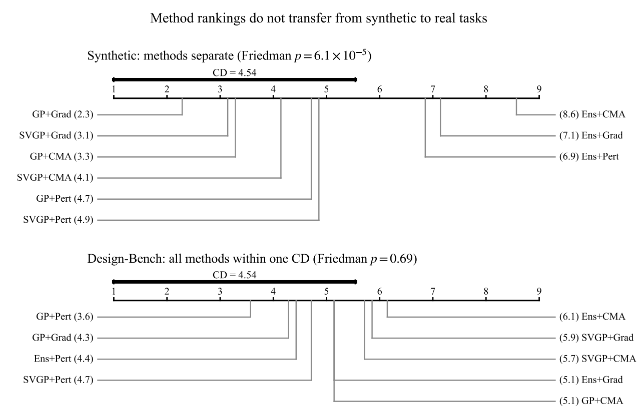

Figure 3: Critical-difference diagram (Nemenyi, α=0.05) over the 9 grid cells. Synthetic tasks separate the cells sharply (Friedman p=6.1×10^{-5}); Design-Bench does not (p=0.69), all cells falling within one critical difference. Method rankings do not transfer from synthetic to real.

A distribution-free repair, and where it transfers. Split-conformal calibration replaces the arbitrary β with a data-driven multiplier.

Proposition 2 (Conformal repair; shift-limited transfer)**.** *Let* $\{(x_i, f(x_i))\}_{i=1}^{n}$ *be exchangeable from* P*,* $r_i = (\mu(x_i) - f(x_i))/\sigma(x_i)$ *the signed one-sided nonconformity, and* $\hat{q}$ *the* $\lceil (n+1)(1-\delta) \rceil$*-th smallest* r_i*. Then for a fresh* $x \sim P$*,* $\Pr(f(x) \geq \mu(x) - \hat{q}\,\sigma(x)) \geq 1-\delta$*. Under a shifted proposal* $\Pi \neq P$ *with density ratio* $w = d\Pi/dP$*, validity is restored*

Table 3: Offline MBO on Design-Bench (16 seeds; normalized 100th-percentile score, higher is better; random-forest oracles for the non-exact tasks). **Bold** = best per task including the COMs/CbAS baselines. Unlike the synthetic grid, the methods cluster tightly and the per-task winner is split across surrogates and optimizers (Friedman p=0.69 over the grid); the ensemble/COMs gradient-ascent collapse persists only on GFP.

Surrogate×Opt.	TF-Bind-8	TF-Bind-10	Superconductor	GFP	UTR	Ant	D'Kitty
Ens × Grad	2.20	1.34	0.99	−9.61	**1.01**	0.92	0.74
Ens × Pert	1.00	1.31	1.29	2.48	0.98	1.29	1.05
Ens × CMA	2.06	1.15	1.18	−5.63	0.99	0.89	0.72
GP × Grad	1.00	1.31	1.24	2.45	0.99	1.52	1.06
GP × Pert	1.00	1.31	1.31	2.45	0.99	1.52	**1.11**
GP × CMA	1.00	1.00	1.26	2.46	0.99	1.52	1.02
SVGP × Grad	1.84	1.11	1.07	1.80	0.99	0.99	1.02
SVGP × Pert	1.45	1.31	1.25	2.45	0.99	1.52	1.09
SVGP × CMA	1.83	1.11	1.26	**2.48**	0.93	0.91	1.02
COMs (baseline)	**2.21**	**1.35**	1.01	−9.20	**1.01**	0.98	0.95
CbAS (baseline)	2.12	1.28	**1.36**	1.85	0.99	**1.53**	1.02

Table 4: Gap controls (synthetic; task-normalized GP − ensemble marginal, task+seed hierarchical bootstrap 95% CIs, B=10,000). The two β rows are read on a single β-invariant normalizer—one per-task min–max fit once over the pooled cell means of *both* conditions—so they sit on one ruler and their difference is interpretable; the paired increment is 0.203 $[0.007, 0.396]$, p=0.020. With pessimism removed entirely, 61% of the gap survives: the edge rests on a mean-quality base that pessimism amplifies. The subsample row is reported on the legacy per-condition normalizer and is *not* comparable to the β rows.

Setting	gap	95% CI
β=2 (pessimism on)	0.525	$[0.406, 0.614]$
β=0 (pessimism off)	0.319	$[0.196, 0.460]$
legacy normalizer—not comparable to the rows above		
ensemble on the GP's 800-pt subsample	0.76	$[0.29, 1.32]$

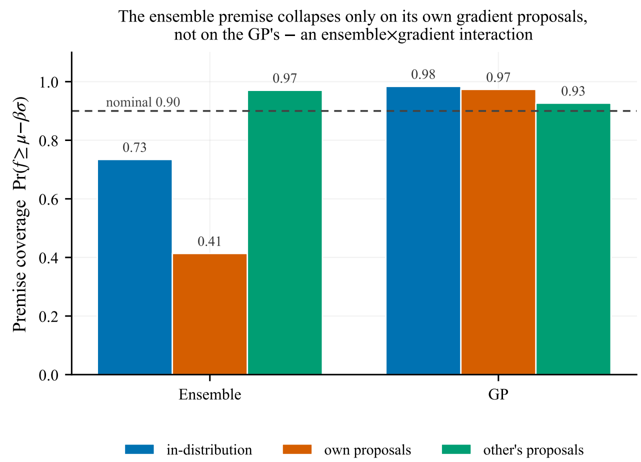

Figure 4: Premise coverage $\Pr(f \geq \mu - \beta\sigma)$ at β=2 (synthetic mean), each surrogate on in-distribution points, its own proposals, and the other surrogate's proposals. The ensemble premise collapses (0.41) *only* on the designs its own gradient ascent returns; on the GP's proposals it is well covered (0.97), and the GP premise holds everywhere. The failure is the ensemble×gradient interaction, not a surrogate defect.

by weighting the calibration quantile by w (weighted conformal; Tibshirani et al. 2019).

Empirically the fitted multiplier varies widely across tasks (synthetic $\hat{q} \in [1.8, 16]$), quantifying how far raw σ undercovers. Replacing $\beta\sigma$ by $\hat{q}\sigma$ restores in-distribution coverage to its 0.90 target on every task (mean 0.90 synthetic and real), exactly as Prop. 2 predicts; but on the shifted proposal Π coverage stays low (0.51 / 0.31 mean, near zero on the sharpest-shift tasks), because $\Pi \neq P$. This yields a concrete recipe (Algorithm 1): a *coverage diagnostic* that reports in-distribution and proposal coverage, flags an unreliable LCB when they diverge, and a *weighted-conformal repair* when a density-ratio estimate is available. The negative transfer is itself the actionable signal: an LCB whose proposal coverage is near zero should not be trusted regardless of β.

7 Discussion, Guidance, and Limitations

Practitioner guidance. (i) On smooth, low-to-moderate-dimensional problems, a GP/SVGP surrogate is robust to the acquisition optimizer (its smooth mean offers no exploitable argmax); a neural ensemble is not, and should be paired with a conservative optimizer (perturbation/CMA-ES) rather than aggressive gradient ascent. (ii) Before trusting an LCB, run the coverage diagnostic (Algorithm 1); if proposal coverage is near zero, the pessimism is not doing what the theory says, and neither β-tuning nor in-distribution conformal calibration fixes it. (iii) On real Design-Bench tasks we detect no difference between methods in this family at standard seed counts, which at n=7 tasks is a statement about power and not a demonstration of equivalence. Note that predictive accuracy is not the lever it appears to be: the ensemble is the *more* accurate surrogate in-distribution and still loses, so effort is better spent on the geometry of the surrogate's mean

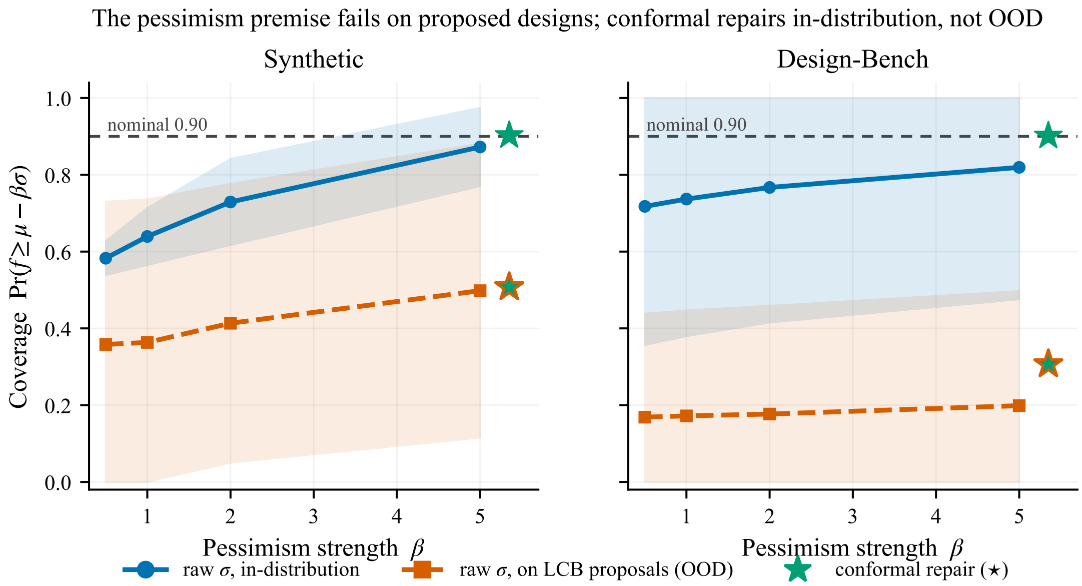

Figure 5: Coverage of the pessimism premise $\mu - f \leq \beta\sigma$ vs. β on synthetic (left) and Design-Bench (right) tasks: in-distribution (solid) and on the optimizer's OOD proposals (dashed), with ± 1 s.d. bands across tasks and the split-conformal repair shown as stars ($\star$). Ensemble in-distribution coverage is moderate, but proposal (OOD) coverage is far lower and task-dependent—near zero on the tasks where gradient ascent collapses; conformal recalibration lifts in-distribution coverage to its 0.90 target (upper star) but leaves the OOD coverage the bound actually needs far below it (lower star).

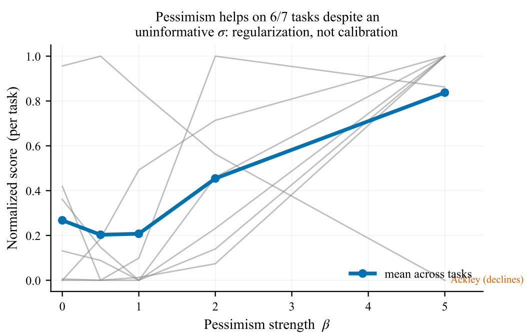

Figure 6: Pessimism sweep: per-task normalized 100th-percentile score vs. β (faint lines, individual synthetic tasks; bold line, mean). Score rises with β on 6 of 7 tasks even though σ is an uninformative error signal ($\rho \approx 0.1$)—pessimism acts as distance-regularization. Ackley is the single task where the penalty points the wrong way.

under optimization pressure than on lowering its held-out error.

Anticipated objections. Three objections deserve direct answers. (i) *That these benchmarks are imperfect is already known.* The contribution is its first controlled measurement—a crossed surrogate×optimizer decomposition with a matched-tuning control, and a premise-coverage frequency not previously reported. (ii) *There is no new state of the art.* The constructive artifact is the coverage diagnostic and weighted-conformal repair (Algorithm 1): a reusable check, the same form of contribution as the evaluation tools

Algorithm 1 Coverage diagnostic and conformal-LCB repair

Input: surrogate (μ, σ), data $\mathcal{D}$, level δ, proposals Π
Output: multiplier $\hat{q}$, coverages $(\hat{c}_{\text{in}}, \hat{c}_{\text{ood}})$, flag

Split $\mathcal{D}$ into fit / calibration folds.
$r_i \leftarrow (\mu(x_i) - f(x_i))/\sigma(x_i)$ on calibration fold (signed, one-sided; floor σ).
$\hat{q} \leftarrow \lceil (n+1)(1-\delta) \rceil$-th smallest r_i.
$\hat{c}_{\text{in}} \leftarrow \widehat{\Pr}_{x \sim \mathcal{D}}(f \geq \mu - \hat{q}\sigma)$.
$\hat{c}_{\text{ood}} \leftarrow \widehat{\Pr}_{x \sim \Pi}(f \geq \mu - \hat{q}\sigma)$.
if $\hat{c}_{\text{ood}} < 1 - \delta$ **then**
    flag LCB unreliable; if $w = d\Pi/dP$ available, reweight r_i by w and refit $\hat{q}$.
end if
return $\hat{q}$, $(\hat{c}_{\text{in}}, \hat{c}_{\text{ood}})$, flag

this line of work is built on. (iii) *The real-task result is null.* It is a significance collapse, not a claim of equivalence—the differences are unresolved, not provably zero (underpowered at $N{=}7$)—a benchmark-validity statement that the rewarded differences do not transfer.

Scope and external validity. The synthetic→real collapse is a statement about the *measured* regime. Two boundary conditions bound it. First, for the non-exact tasks we substitute a random-forest oracle to remove simulator and framework dependence—a genuine substitution on GFP, UTR, Ant, and D'Kitty (the native oracle is an RF only for Superconductor), so a different oracle could re-separate methods. But it does not manufacture the null: on the three tasks with no smoothing RF substitution (TF-Bind-8/10 exact, Superconductor native RF), the omnibus stays flat (Friedman $p{=}0.93$) and the median 9-cell spread (0.34) is no smaller than on the substituted tasks (0.39). Second, the real-task omnibus is over 7 tasks, and the equivalence test is underpowered there (Section 4): we can say the method differences are unresolved, not that they are zero.

Limitations. Our real-task evidence is Design-Bench with random-forest oracles for the non-exact tasks; the null is scoped to that suite and to CPU-scale training. A cross-check ran official COMs and CbAS: our CbAS matches the official scores ($|\Delta|{=}0.004$ on TF-Bind-8), while our COMs and official COMs diverge ($|\Delta|{=}1.2$, a reduced-epoch official run and a different oracle variant), so we report both side by side. The synthetic datasets are fixed across seeds (seed-0), so reported variance is training/optimization, not data-draw. Premise coverage on the discrete tasks uses relaxed logits, not decoded sequences, so those numbers (notably GFP) are softer—we lean the coverage mechanism on the synthetic tasks. Weighted conformal needs a density-ratio estimate we do not learn; we characterize the transfer gap rather than close it.

8 Conclusion

The reported low-dimensional advantage of GP-LCB in offline MBO is not a clean surrogate-class effect and not a calibration effect: it is a surrogate$\times$optimizer interaction driven by the GP's smooth posterior mean, on which gradient ascent finds no over-estimated argmax to exploit as it does on the ensemble's jagged mean. Removing pessimism ($\beta{=}0$) leaves the edge intact and giving the ensemble the GP's data does not close it—the cause is inductive bias, not uncertainty. Making the premise measurable (coverage equals LCB validity) turns it into a diagnostic: a bound whose proposal coverage is near zero should not be trusted regardless of β. And the differences these synthetic benchmarks reward are unresolved on real tasks—the same prior–task match, seen from both sides. We contribute the decomposition, the mechanism and its controls, the coverage diagnostic, and the synthetic-to-real validity result.

Ethical Statement

This work studies optimization methodology on public benchmarks and introduces no new data or deployed system. By flagging when a pessimistic bound is invalid, the coverage diagnostic aims to reduce over-confident designs downstream.

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